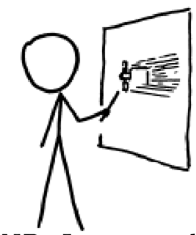
PH-429 Chaos and Nonlinear Dynamics

-

Engineering School, Physics Department

Fall 2026

Meetings: Tuesdays 9 am - 12 noon room 803, 41 Cooper Sq



Philip Yecko (feel free to call me Phil)
philip.yecko@cooper.edu, payecko.physics@gmail.com
Office: 41 CSq Room 515
Office Hours: see MOODLE including Zoom link
iPhone: (862) 485-3557

NB: I reserve the right to make changes to this syllabus; all changes will be announced and posted.

Course Description: Course examines the mathematical formalism and methods of analysis and solution of deterministic autonomous and non-autonomous nonlinear systems, including phase space, local and global bifurcations, limit cycles and attractors, strange attractors, chaos and fractals, maps and universality. Applications will be drawn from nonlinear oscillators in biology, chemistry and electronics, mechanical and structural stability, geophysics, astrophysics and fluid dynamics, as well as from models used in climate science, economics and epidemics.

Prerequisite(s): Ph 214, Ma 113 (Ma 240 preferred) and CS 102 or ECE 160

Credit Hours: 3 (expect to invest ∼3-4 times this number of hours per week of this course)

Text(s): Required: Steven H. Strogatz *Chaos and Nonlinear Dynamics* SHS or just "Strogatz"
Supplemental: Selections from P. Manneville *Dissipative Structures and Weak Turbulence*; J. Guckenheimer & P. Holmes *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields* plus other texts and journal articles.

Course Objectives: You should finish the course understanding the origin, nature and impact of nonlinearity in mathematical models –generally systems of nonlinear differential equations– and their solutions; how to formulate and analyze such models mathematically, using both direct and indirect methods, including but not limited to: linearization and eigenspace analysis and bifurcations and relevant mathematical theorems; phase plane and phase space analysis; existence and behavior of special solutions such as limit cycles and attractors, including strange attractors; role of dimensionality, geometry and topology in phase flows and solutions, including fractional dimensions; applications in biology, chemistry, engineering (including fluid dynamics), epidemiology, geophysics (including weather and climate) and physics will be treated throughout, both from a modeling and data analysis perspective, including computational considerations and methods. Some advanced topics will also be addressed, including: role of dissipation vs Hamiltonian systems; global bifurcations; projection, dimension reduction and normal forms; the relation of the chaotic systems of this course to spatio-temporal systems involving patterns. [ABET outcome criteria: 1,2,4,6 & 7]

Assessments and their weights for grading:

Problem Solving	50%
Final Project or Exam	50%

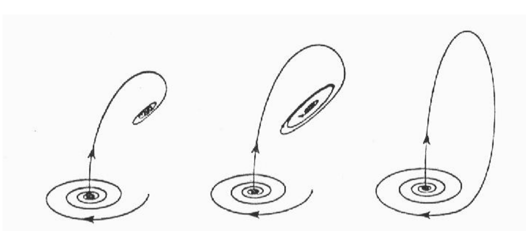
Policy on Assessments: Students will be assessed based on their grasp of the material as presented and discussed in weekly meetings, which will include discussion of both concepts and solutions of problems, some given in advance as problem sets, others uncovered *in medias res.* A final project will be developed iteratively and due no later than the final week of the semester.

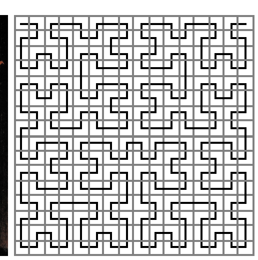
Letter Grades: A,B,C,D and F grades are assigned based on above weighted scheme.

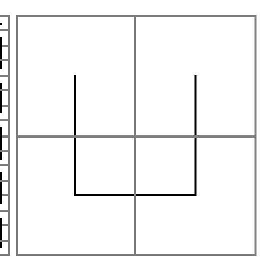
Outline of Topics (tentative, may be adjusted according to the progress of the course) :

Weeks 1-3 • THE DISCRETE: 1D Maps, Logistic Equation and Map, "Period Three means Chaos," Lyapunov Exponent, Universality, Countable and Unvountable Sets, Cantor Sets, Dimension, Self-Similarity, Fractals (SHS chs. 10,11)
Weeks 4-6 • THE ONE-DIMENSIONAL: Flows in 1D, Fixed Points and Stability, Populations, Potentials, Bifurcations (saddle-node, transcritical, pitchfork), Catastrophes, Periodic Flows, Oscillators, Pendula and Rufus T. Firefly (SHS chs. 2,3,4)
Weeks 7-9 • THE TWO-DIMENSIONAL: Types of Linear Systems, Love Affairs, Phase Plane & Phase Portraits, Topology and Theorems, Fixed Points and Linearization, Foxes, Rabbits and other barnyard creatures (SHS chs. 5,6)
Weeks 10-11 • OSCILLATIONS: Limit Cycles, Poincare-Bendixson, Relaxation Oscillations, Hopf Bifurcations, Global Bifurcations, Poincare Maps (SHS chs. 7,8)
Weeks 12-14 • THREE (OR MORE) DIMENSIONS AND CHAOS: Lorenz Systems, Strange Attractors and Chaos, Lorenz Map, Henon Map, Other Systems- Rossler, Moore-Spiegel, Forcing and "Whose Water-wheel was it?" (SHS ch. 9)
Week 15 • ADVANCED TOPICS: TBD

Course Policies: As an Independent Study Course, please simply refer to Cooper Union's general policies (links below) on any and all matters such as academic honesty, counseling and mental health, accommodations, disabilities and other matters of civility, including any form of discrimination including but not limited to those covered by federal civil rights Titles VI, VII and IX plus Section 504; Cooper Union policy on sexual misconduct (Title IX) can be found here. The ABET/NSPE Codes of Ethics (link here), Cooper Union's academic integrity and standards: found here, counseling and mental health services linked here, accommodations and support services for students are described here, and the process described here. (PDF includes active links)







Sensitive Systems Introduction to This Course by E.A. Spiegel and P.A. Yecko

The word *gas*, which derives from the Greek word *chaos*, was introduced into scientific use by Jan Baptista van Helmont[1] a seventeenth-century Dutch scientist and alchemist. The sense of the Greek word is to be found in Greek mythology, the story of Chaos and her offspring. "Hesiod's *Theogony* describes the origins of [the cosmos]. This begins with the Greek name Chaos having the connotation of *gap*."[2] Since gases are rarefied media, the Greek sense of emptiness seems appropriate to their study. "In English, of course, chaos has come to mean confusion and disorder which for scientists, have not negative connotation."[1] The scientific study of such things is what this course is about.

In science, the word *chaos* was used quite naturally in speaking of early studies of the formation of the universe. The word *chaology* is to be found in the OED and it evokes the subject, as we view it even now, quite vividly:

> *Then Zeus no longer held back his might*;
> *flame unspeakable rose .... Astounding heat seized*
> *Chaos ...*[3]

The creation of the universe in Western religious tradition also refers to a primordial disorder, the "*unformed and void*."[4] In recent years, chaos has come to mean something somewhat different, more subdued, but it remains a fascinating topic for all that. One aim of this course is to convince you of that.

These words *chaos* and *turbulence*, it's echo in the study of fluids, are hard to define and scientists are not agreed about the definitions they should offer. This is not a new situation. St. Augustine said: "I know what time is until someone asks me to define it." And early on in his book on Money, J.K. Galbraith says "It will be asked in this connection if a book on the history of money should not begin with some definition of what money really is .... The precedents for such effort are not encouraging. Television interviewers with a reputation for penetrating thought regularly begin interviews with economists with the question: 'Now tell me, just what is money anyway.' The answers are invariably incoherent.[5]

Though the present interest in the subject of chaos may be partly explained by its name, which resonates through millennia, and even though it has now been given a new sense it is still associated with the the notion of disorder. But the idea of uncertainty and disorder is not a new one, even though there is some novelty in the modern use of the word. And that idea is already seen in Bernoulli's *The Art of Guessing* when he says[6]

> *... who could enumerate the countless changes that the atmosphere*
> *undergoes every day, and from that predict today what the weather*
> *will be a month from or even a year f rom now? ... These and similar*
> *forecasts depend on factors that are completely obscure, and which*
> *constantly deceive our senses by the endless complexity of their*
> *interrelationships, so that it would be quite pointless to proceed along this road.*

There is no surprise here; the notion expressed by Bernoulli is a commonplace of science fiction stories and is expressed in many an adage. As Erasmus of Rotterdam said, speaking of something else: "Almost everyone knows this, but it has not occurred to everyone." (He could not have said it of Bernoulli's remark since *Ars Conjectandi* was written well after the fifteenth century in which Erasmus lived.) Already for decades, fluid dynamicists discussing the subject of fluid turbulence (a word that was adopted by physical scientists only in the previous century) have repeated the trope:

"small causes produce large effects." This bit of folk wisdom is an almost direct translation of an old Dutch proverb that we suspect came into science by way of Jan Burgers, who was an energetic student of turbulence fifty years ago. Whatever the truth of this attribution may be, the aphorism does describe the mechanism behind what is now called chaotic behavior.

One of the reasons that chaotic systems have proved interesting is that, though they may look simple, their behavior can be very complicated, whereas previously it had been thought that to obtain complexity we would have to study complex systems. The prospect of extracting complicated behavior from simple systems offers the hope that we may be able to understand complicated behavior at some level. Phenomena that were once thought to be inaccessible to most scientists can now be studied by anyone who knows a bit about coding. Not long ago, most scientists thought that to study highly disordered systems they would have to use complicated models or statistical methods that are based on the notion that the rules of behavior are not deterministic. But the discovery that disorder could appear in relatively simple causal systems systems whose behavior is completely governed by explicit and well-defined rules made people realize that they could get rich behavior from simple looking systems.

We have yet to clarify what we mean by *disorder* but the general intuitive idea of a behavior that is not periodic will serve for now; words like complexity and irregularity are also used in this context. The point though was that for the past 75 years, with increasing frequency, people were finding model systems for erratic variations in time or for complicated spatial structures that were in themselves very simple. That is they seem to fit into the picture that has been considered ideal from the time of Newton of having simple and explicit rules of behavior, rules that tell you what a system will do next if you knew what its state is now. Such systems are called deterministic and these had been expected to conform to the view of science expressed by Laplace in 1820:

An intellect which at a given instant knew all the forces acting in nature and the positions of all things of which the world consists - supposing the said intellect were vast enough to subject these data to analysis - would embrace in the same formula the motions of the greatest bodies in the universe and those of the slightest atoms; nothing would be uncertain for it, and the future, like the past, would be present to its eyes.

But a surprise was in store. In systems with simple behavior such as periodic pendula in certain cases, if you make a slight error in specifying the state of the system and then proceed to compute its state for later times, you find that the error grows linearly in time, in general. But for systems exhibiting chaotic behavior, the initial error grows exponentially. This means that very soon the error has gotten so large that no reasonable prediction about the future behavior of the system can be made. This seems simple enough, but it represents a great change in our expectations of what can be done by scientific methods. The implications of this realization is another thing that this course is about.

Given how simply stated the point is, it may seem surprising that the Laplacian view of science was largely accepted, although it was often put aside out of practical necessity when dealing with the complicated behavior of highly complex systems, like the ones Bernoulli had in mind. That is, methods of probability theory were introduced when speaking of the outcome of a coin toss (*Is it?*) or the behavior of a gaseous medium. (There are also issues about the probabilistic interpretation of quantum mechanics that we will likely not have time to go into.) Few people took much note of Poincare's realization that anything short of Laplace's remarkable being would have serious trouble with the the three-body gravitational problem, Einstein being a notable exception. Though most people nowadays recall that even simple looking systems can exhibit complex behavior, the distinction formerly made in this respect is still there in some minds.

Not so long ago, the cosmologist Hermann Bondi wrote that in his view, ‘most of science is not like the Solar System but much more like weather-forecasting.’ In fact, a better picture would be that these systems do not differ so much qualitatively –they are both sensitive systems– but quantitatively in that the numbers of their degrees of freedom are rather different, and that is the technicality that makes our ways of handling them different.

In this course, we shall examine systems that behave chaotically and seek the origins of this behavior. We shall be interested in understanding why nature looks complicated but, as we learn over and over again, is basically simple. Admittedly, simplicity may be largely in the eye of the beholder, but if you look long enough at the objects of this course, you will begin to see it. Our goal is to get some idea of how complexity arises and how it may be described. But a higher goal for you would be to go beyond this first step, now or later, and see how the models used in this subject can be added to the arsenal of mental tricks you accumulate to attack new problems of your own. To get to the point where you can do this, you will need to learn some fundamentals and some mathematical concepts. As you will see, model making remains an art and we shall not here be able to turn it into a science. It will be up to you to take what you can from this study and go beyond it. So while this course may be, to some extent, akin to a science appreciation course, we hope that it may also be more for some of you – a real science course. In any case, there is no doubt much benefit to be derived from seeing how the theory of chaos has been used to understand many complicated problems.

---

1. R. Hoffmann and V. Torrence, The Sciences, Nov.-Dec. 1993, p.29
2. G.E.R. Lloyd, Greek Cosmologies in ’Ancient Cosmologies’ edited by Carmen Blackes & Michael Loewe (George Allen and Unwin Ltd, London) p. 200, 1975
3. Hesiod, ”The Homeric hymns and Homcrica,” Translated by H.G. White (Harvard University Press, Cambridge Mass.) p. 87.
4. The Bible; Genesis 1:2 from Hebrew (MT): *tohu v’bohu*; same verse is similarly translated from the Greek Septuagint (LXX) as “unformed and void.”
5. .J. K. Galbraith, Money (Houghton Mifflin Co., Boston) p. 5, 1975.
6. Kasner and Newman, The World of Mathematics, vol. 3, p. 1453 ff.